How LLMs Violate Ontology Constraints: A Landscape Study and Grammar-Based Mitigation

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Abstract

Large language models (LLMs) used for zero-shot information extraction (IE) systematically violate ontology type constraints—producing entity types outside the schema vocabulary, hallucinating relation names, and generating domain/range-incompatible triples. Yet no prior work has systematically measured *how* and *where* these violations arise across ontologies of varying complexity. We present three contributions. **(1)** A *violation landscape study* spanning two IE benchmarks (SciERC, 7 relations; DocRED, 96 relations) and two LLMs (Qwen3-8B, Llama-3.1-8B), revealing that domain/range violations concentrate on specific relation types and that constraint tightness predicts violation rate in small ontologies (ρ=0.85) but loses predictive power in complex ones (ρ=0.14, p=0.18). **(2)** A grammar-constrained decoding layer compiled automatically from ontology definitions that *universally eliminates* entity-type and relation-vocabulary violations (ET→0%, RV→0%) at only 14% latency overhead and zero F1 cost across both datasets. **(3)** A semantic constraint layer providing per-relation domain/range enforcement, which achieves 94% domain/range violation reduction on a tight ontology (SciERC) but only 35.5% on a complex ontology (DocRED)—an empirical result we report transparently, identifying scaling semantic constraints to complex ontologies as an open problem. All experiments use zero-shot extraction to isolate constraint enforcement effects from task-specific training.

1 Introduction

LLM-based information extraction has become a practical paradigm for populating knowledge bases without task-specific training data [Wei et al., 2023]. A user provides an ontology schema—entity types and relation definitions with domain/range constraints—and the LLM generates structured output in a single pass. However, generated triples frequently violate the ontology constraints they are supposed to respect: entity types fall outside the schema vocabulary, relation names are hallucinated, and argument types violate domain/range specifications.

These violations matter. A triple (`Washington [ORG], P131_located_in, United States [LOC]`) may be factually correct, but if the ontology specifies that `P131`'s subject must be `LOC`, the `ORG` typing is structurally invalid and will be rejected by downstream systems that enforce schema compliance. In safety-critical domains (biomedical, legal, regulatory), even statistical reduction of violations is insufficient—per-instance compliance is required.

Despite the practical importance of this problem, no prior work has systematically characterized *which* violations occur, *where* they concentrate, and *how* their distribution changes with ontology complexity. We address this gap with three contributions:

Contribution 1: Violation Landscape Study. We introduce a four-dimensional violation taxonomy (entity type, relation vocabulary, domain/range, cross-field consistency) and measure violation rates on SciERC (7 relations) and DocRED (96 relations) with Qwen3-8B and Llama-3.1-8B. We find that violations are not uniformly distributed—they concentrate on specific relation types—and that constraint tightness (the fraction of valid entity-type pairs for a relation) is a strong predictor of violation rate in small, well-defined ontologies (SciERC, ρ=0.85)

but loses predictive power in complex ontologies (DocRED, ρ=0.14, p=0.18), where model priors, data frequency, and entity-type granularity dominate.

Contribution 2: Grammar-Constrained Decoding as Universal Structural Validity Layer. We compile ontology schemas into context-free grammars (via XGrammar) that constrain autoregressive decoding. This eliminates entity-type violations (ET→0%) and relation-vocabulary violations (RV→0%) with only 14% latency overhead and no degradation in extraction F1. The grammar layer generalizes perfectly across both datasets, requiring no dataset-specific tuning.

Contribution 3: Semantic Constraint Layer as Proof-of-Concept. We implement a stateful semantic constraint layer that enforces per-relation domain/range rules via dynamic logit masking during decoding. On SciERC (7 tight relations), this achieves 94% domain/range violation reduction. On DocRED (96 loosely-constrained relations), reduction is 35.5%. We explicitly identify scaling semantic constraints to complex ontologies as an open problem, reporting the DocRED residual violation rate (10.19%) as a quantified challenge for future work.

2 Experimental Setup

2.1 Datasets and Ontologies

We evaluate on two IE benchmarks with contrasting ontology characteristics:

SciERC [Luan et al., 2018] contains 551 scientific abstracts with 7 relation types. The ontology has tight domain/range constraints: each relation permits only 2–23 of the 36 possible entity-type pairs (6 entity types × 6 types).

DocRED [Yao et al., 2019] contains 998 documents with 96 Wikidata relation types and 4 coarse entity types (PER, ORG, LOC, MISC). Despite the larger relation vocabulary, individual constraints are looser: high-frequency relations (e.g., `P131_located_in`) permit most type combinations.

2.2 Models

Our primary model is **Qwen3-8B-Instruct**, evaluated in zero-shot mode with a minimal prompt containing only the task description and output format. For the cross-model analysis (§3.3), we additionally evaluate **Llama-3.1-8B-Instruct** under identical conditions.

2.3 Violation Taxonomy

We define four violation types with increasing semantic complexity:

1. **Entity Type (ET)**: Generated entity type $\notin$ schema-defined type vocabulary.
2. **Relation Vocabulary (RV)**: Generated relation name $\notin$ schema-defined relation set.
3. **Domain/Range (DR)**: Entity types of a triple's head/tail violate the relation's domain/range specification.
4. **Cross-Field Consistency (CF)**: Same entity receives different type labels across different triples (not a per-triple violation but a global consistency measure).

ET and RV are *structural* violations expressible as context-free constraints. DR requires cross-field state (knowing the relation to check argument types). CF is a global property not enforceable by per-triple decoding.

Table 1: Unconstrained violation rates (Qwen3-8B, zero-shot). DR is the dominant violation type on both datasets. RV is low on SciERC (7 natural-language relation names) but high on DocRED (96 Wikidata property IDs).

Dataset	Rels	DR(%)	RV(%)	ET(%)	R-F1(%)	Triples
SciERC	7	15.4	1.39	0.4	8.23	1,724
DocRED	96	15.8	26.5	6.8	8.08	16,200

Table 2: DocRED per-relation DR by constraint tightness quartile. Unlike SciERC (ρ=0.85), tightness does not predict violation rate on DocRED (ρ=0.14, p=0.18).

Quartile	N	Med. Triples	Mean DR(%)	Near-Zero Frac.
Q1 (tightest)	24	42	12.8	0.33
Q2	24	85	9.5	0.42
Q3	24	120	10.2	0.38
Q4 (loosest)	24	210	8.9	0.42

3 Violation Landscape Study

This section presents our primary empirical contribution: a systematic measurement of how LLMs violate ontology constraints across datasets of varying complexity.

3.1 Violation Rates Across Ontology Complexity

Table 1 shows unconstrained violation rates on both datasets using Qwen3-8B.

Two observations emerge. First, domain/range violation rates are comparable across a 14× ontology size difference (15.4% vs. 15.8%), suggesting that *violation severity is not driven by ontology size alone*. Second, the violation *profile* differs markedly: SciERC exhibits low RV (1.39%) because the model can produce natural-language relation names reliably, while DocRED's Wikidata property IDs (`P17`, `P131`, etc.) yield 26.5% RV—the model frequently generates plausible but invalid property IDs.

3.2 Per-Relation Violation Distribution and Constraint Tightness

Violations are not uniformly distributed across relations. On SciERC, `Compare` exhibits 42.4% DR while `Used-for` shows only 0.76%. We define *constraint tightness* as the ratio of valid entity-type pairs to all possible pairs for a given relation. On SciERC, tighter constraints correlate strongly with higher violation rates (Spearman ρ=0.85): relations permitting few type combinations are precisely those the model violates most.

Tightness as predictor: SciERC vs. DocRED. Table 2 shows that this relationship breaks down on DocRED. The Spearman correlation between constraint tightness and per-relation DR drops to ρ=0.14 (p=0.18, not statistically significant). Mean DR varies only weakly across tightness quartiles (8.9%–12.8%), with 38% of relations at near-zero DR regardless of tightness.

Table 3 illustrates why. Tight relations can have low DR when the model's prior aligns with the constraint (`P35_head_of_state`: tightness=2 valid pairs, DR=2.2%). Loose relations can have high DR when entity-type boundaries are ambiguous (`P131_located_in`: 15 valid pairs, DR=14.8% due to ORG/LOC/MISC confusion).

Implication. Constraint tightness is a useful predictor in small, well-defined ontologies but loses power in complex ontologies where model priors, data frequency, and entity-type granu-

Table 3: Counter-examples to the tightness→DR hypothesis on DocRED.

Relation	Tight.	Valid Pairs	Triples	DR(%)	Explanation
P35_head_of_state	tight	2	45	2.2	Model prior aligns with constraint
P131_located_in	loose	15	580	14.8	LOC/ORG/MISC boundary confusion
P710_participant	tight	3	120	18.3	Model assigns diverse types to participants
P27_citizenship	loose	12	95	1.1	Strong PER→LOC prior

Table 4: Cross-model violation comparison on SciERC. Absolute rates differ, but per-relation rank order is preserved (ρ=0.82).

Model	DR(%)	RV(%)	ET(%)	Parse(%)	R-F1(%)	Triples
Qwen3-8B	15.4	1.39	0.4	100	8.23	1,724
Llama-3.1-8B	9.8	3.5	1.2	92	6.5	1,580

larity interact in ways not captured by a single structural measure. This finding has practical consequences for constraint enforcement: approaches that prioritize tight constraints (expecting them to be most-violated) will misallocate effort on complex ontologies.

3.3 Cross-Model Consistency

To test whether the violation landscape is model-specific, we evaluate Llama-3.1-8B-Instruct on SciERC under identical conditions (Table 4).

Absolute violation rates differ substantially (DR: 15.4% vs. 9.8%), but the per-relation rank order is highly consistent. Table 5 shows that the same relations are most-violated and least-violated across both models, with Spearman rank correlation ρ=0.82.

This suggests that the *relative structure* of the violation landscape—which relations are hard vs. easy—is a property of the ontology and task, not the specific model. We note two caveats: Llama's lower parse rate (92% vs. 100%) introduces survivorship bias (violations in unparseable outputs are not counted), and the absolute rate difference means problem severity is partly model-dependent.

4 Grammar-Constrained Decoding for Structural Validity

The violation landscape study reveals two violation types—ET and RV—that are purely structural: they depend only on whether the generated token falls within a fixed vocabulary. These are naturally expressible as context-free grammar (CFG) constraints.

4.1 Ontology-to-Grammar Compilation

Given an IE ontology schema defining entity types $\mathcal{T}$ and relation names $\mathcal{R}$, we automatically compile a JSON-structure grammar that constrains the output to:

- Entity types $\in \mathcal{T}$ at entity-type positions
- Relation names $\in \mathcal{R}$ at relation-name positions
- Valid JSON structure throughout

Table 5: Per-relation DR rank comparison (SciERC). `Compare` and `Feature-of` are top violators for both models (ρ=0.82).

Relation	Qwen DR(%)	Rank	Llama DR(%)	Rank
Compare	42.4	1	31.0	1
Feature-of	33.5	2	18.0	3
Part-of	25.0	3	15.0	4
Evaluate-for	24.1	4	19.0	2
Hyponym-of	14.5	5	11.0	5
Used-for	0.76	6	0.5	6
Conjunction	0.0	7	1.0	7

Table 6: Grammar-constrained decoding eliminates ET and RV violations universally, at zero F1 cost.

Dataset	Mode	ET(%)	RV(%)	DR(%)	R-F1(%)	Overhead(%)
SciERC	Unconstrained	0.4	1.39	15.4	8.23	—
	Grammar-only	0.0	0.0	—	—	14.3
DocRED	Unconstrained	6.8	26.5	15.8	8.08	—
	Grammar-only	0.0	0.03	26.17	8.24	14.6

We implement this using XGrammar, which compiles CFGs into finite-state machines (FSMs) for efficient per-token masking during autoregressive decoding. Grammar compilation is a one-time cost (0.8s) amortized across all documents.

4.2 Results: Universal Elimination of ET and RV

Table 6 shows grammar-constrained decoding results on both datasets.

ET violations drop to exactly 0% on both datasets. RV drops to 0% on SciERC and 0.03% on DocRED (a single residual instance). F1 is preserved: DocRED grammar-only R-F1 (8.24%) is slightly above unconstrained (8.08%). Latency overhead is modest at 14.3–14.6%, because XGrammar's FSM-based masking uses efficient bitwise operations on GPU.

DR increases under grammar-only on DocRED. An initially surprising result: grammar-only DR on DocRED is 26.17%, *higher* than unconstrained DR (15.8%). This is an **artifact of measurement, not degradation**. Without grammar constraints, 26.5% of triples have invalid relation names (RV violations). These triples cannot be evaluated for DR because their relation is not in the ontology—the DR checker skips them. Grammar-only decoding forces all relations into the valid vocabulary, exposing domain/range violations that were previously masked by RV violations. The 26.17% grammar-only DR is a more accurate estimate of the true DR rate than the unconstrained 15.8%, which systematically undercounts by excluding RV-violating triples from DR evaluation.

4.3 Latency Analysis

Table 7 reports wall-clock latency for all decoding modes on SciERC (single RTX PRO 6000, batch size 1).

On DocRED, grammar-only overhead is comparable (14.6%); the combined grammar+semantic overhead is 54.1% (28.5 vs. 18.5 sec/doc). The semantic layer's overhead (39%) is dominated by Python-level mask computation (entity registry lookups, per-relation constraint table queries), suggesting room for optimization via C++/CUDA kernels.

Table 7: Latency comparison (SciERC, RTX PRO 6000, batch=1). Grammar-only adds 14% overhead; the semantic layer adds 39%; combined is 54%.

Mode	sec/sample	Overhead(%)	tok/s
Unconstrained	2.8	—	42.5
Grammar-only	3.2	14.3	37.2
Semantic-only	3.9	39.3	30.5
Full (G+S)	4.3	53.6	27.8

Table 8: Constraint ablation on SciERC (7 relations). Grammar+Semantic achieves 94% DR reduction at zero F1 cost. Semantic-only achieves only 24% DR reduction due to enabling dependency.

Mode	DR(%)	CF(%)	R-F1(%)	Enabling
Unconstrained	15.4	—	8.23	—
Semantic-only	11.65	—	—	11.65/0.93=12.5×
Grammar+Semantic	0.93	9.9	8.36	(reference)

5 Semantic Constraint Layer: Proof-of-Concept

Grammar constraints eliminate structural violations but cannot enforce domain/range rules, which require cross-field state (knowing the current relation to determine valid argument types). We implement a **TypeConstraintProcessor** that maintains a stateful entity registry during decoding and applies dynamic logit masking at entity-type token positions based on the relation's domain/range specification.

5.1 Architecture

At each entity-type generation step, the semantic layer:

1. Identifies the current relation context from the partially generated output.
2. Looks up the relation's valid (head-type, tail-type) pairs from the ontology.
3. Determines whether the current position is head or tail argument.
4. Masks all entity types that would violate the domain/range constraint.

The semantic mask is composed with the grammar mask via bitwise AND: $M_{\text{final}} = M_{\text{grammar}} \cap M_{\text{semantic}}$. When the intersection is empty (the semantic layer would reject all grammar-valid tokens), the system falls back to grammar-only constraints, maintaining structural validity as a guaranteed floor.

5.2 Results on Tight Ontology (SciERC)

Table 8 shows the full ablation on SciERC.

The combined grammar+semantic system reduces DR from 15.4% to 0.93% (94% reduction) with F1 preserved (8.36% vs. 8.23%, difference not statistically significant).

Enabling dependency. Semantic-only decoding achieves only 24% DR reduction (15.4%→11.65%). The grammar layer is a *prerequisite* for effective semantic enforcement on SciERC: without grammar, 1.39% of triples have invalid relation names, causing the semantic layer to skip constraint

Table 9: Constraint ablation on DocRED (96 relations). Grammar eliminates ET/RV universally. Semantic layer achieves limited DR reduction (35.5%). The 10.19% residual DR remains an open problem.

Mode	DR(%)	RV(%)	ET(%)	CF(%)	R-F1(%)	Triples
Unconstrained	15.8	26.5	6.8	0.1	8.08	16,200
Grammar-only	26.17	0.03	0.0	0.08	8.24	17,341
Semantic-only	11.2	24.8	5.9	3.8	7.95	16,050
Grammar+Semantic	10.19	0.03	0.0	2.99	8.17	17,372

enforcement for those triples. The enabling ratio (semantic-only DR / full DR = 12.5×) quantifies this: the semantic layer is 12.5× more effective when grammar guarantees valid relation names.

5.3 Results on Complex Ontology (DocRED)

Table 9 shows the full ablation on DocRED.

The grammar layer again achieves universal ET/RV elimination. However, the semantic layer's DR reduction is markedly weaker: 35.5% (15.8%→10.19%) vs. 94% on SciERC. The 10.19% residual DR represents 1,770 triples with structurally valid but domain/range-incompatible type assignments.

Enabling dependency vanishes. The enabling ratio drops from 12.5× (SciERC) to 1.1× (DocRED): semantic-only DR (11.2%) is nearly identical to full DR (10.19%). On DocRED, grammar and semantic layers operate as independent contributions rather than a prerequisite chain. This occurs because DocRED uses short Wikidata property IDs (P17, P131, etc.)—even without grammar constraints, most generated relation IDs happen to match valid properties from the 96-relation vocabulary, allowing the semantic layer to look up and enforce constraints regardless.

Why semantic enforcement is weaker on DocRED. Three factors contribute to the limited DR reduction: (1) DocRED's 4 coarse entity types (PER, ORG, LOC, MISC) create inherently ambiguous boundaries—geopolitical entities are simultaneously ORG and LOC—limiting the semantic layer's ability to enforce type distinctions that are not ontologically clean. (2) High-frequency relations (e.g., `P131_located_in`, 580 triples) have loose constraints permitting most type combinations, leaving little room for enforcement. (3) Low-frequency relations (37 with zero DR) are often inherently easy (strong model priors), inflating the "near-zero" fraction without reflecting systematic method success.

5.4 Cross-Field Consistency Side Effects

Both grammar+semantic configurations introduce cross-field consistency (CF) violations absent in unconstrained decoding: 9.9% on SciERC, 2.99% on DocRED. Grammar-only CF is negligible (0.08%), confirming that CF violations are introduced entirely by the semantic layer's per-relation independent type enforcement.

CF sub-type analysis (DocRED). Table 10 decomposes the 519 CF violations on DocRED.

The dominant pattern is **EVR** (entity-vs-relation type drift): the same entity receives different types in the entity list vs. relation triples. Top confusion patterns are ORG↔LOC (95 instances, soft boundary for geopolitical entities), PER↔ORG (72, hard NER boundary), and MISC↔LOC (48, political entities). Entity duplication (85 instances)—where the semantic

Table 10: Cross-field consistency violation breakdown (DocRED, Grammar+Semantic). EVR dominates; entity duplication is a newly identified side effect.

Sub-type	**Count**	**Rate(%)**	**Severity**
EVR (entity-vs-relation drift)	380	1.88	Medium-High
Within-relation inconsistency	45	0.26	High
Entity duplication	85	0.42	Medium
Hallucinated extra triples	∼35	0.20	High

layer's per-relation enforcement causes the model to generate duplicate entities with different type labels—is a newly identified side effect not observed on SciERC.

Limitation. CF violations are an inherent consequence of per-relation independent type enforcement. Each triple individually satisfies domain/range constraints, but cross-triple entity-type consistency is not maintained. The PER↔ORG confusion pattern (72 instances) represents genuine errors that downstream systems cannot dismiss as type-system ambiguity. A global entity-type consensus mechanism (enforcing consistent typing before per-relation enforcement) is a natural direction for future work.

6 Discussion

Grammar constraints as a universal no-regret layer. The grammar layer represents the clearest practical finding: ET→0% and RV→0% across both datasets, at 14% overhead and zero F1 cost. Any LLM-based IE pipeline can add grammar-constrained decoding compiled from its ontology schema with no dataset-specific tuning and no quality trade-off.

DocRED residual DR: an open problem. The 10.19% residual DR on DocRED after full grammar+semantic enforcement is a quantified open problem. Unlike SciERC (0.93% residual), DocRED's coarse type system and loose per-relation constraints fundamentally limit what per-triple decoding-time enforcement can achieve. The tightness-DR null result (ρ=0.14) further implies that addressing this gap requires going beyond structural constraint measures—perhaps incorporating corpus-level type frequency priors or entity-level global reasoning.

Scaling semantic constraints remains open. Our semantic layer demonstrates strong results on a tight, small ontology (SciERC: 94% DR reduction) but limited effectiveness on a complex, loosely-constrained one (DocRED: 35.5%). We explicitly mark this as an open problem rather than claiming general-purpose semantic constraint enforcement. The enabling dependency—strong on SciERC (12.5×), absent on DocRED (1.1×)—suggests that the interaction between grammar and semantic layers is modulated by ontology vocabulary design, not a universal mechanism.

Violation landscape data as a reusable resource. Our per-relation violation measurements, cross-model rank correlations, and tightness analyses provide a reusable empirical foundation for future work on constraint enforcement, error analysis, and ontology design. The finding that model priors interact with ontology structure in complex ways (Table 3) has implications beyond constrained decoding—for prompt design, ontology engineering, and evaluation methodology.

7 Related Work

Grammar-constrained decoding. XGrammar [Dong et al., 2024] and related systems compile formal grammars into per-token masks for autoregressive decoding. Prior applications focus on code generation and structured output formatting. We apply grammar-constrained decoding to IE ontology compliance—a setting where the grammar encodes domain-specific type vocabularies rather than general programming language syntax.

Constrained text generation. NeuroLogic decoding [Lu et al., 2022] enforces lexical constraints; FLAN-based approaches use instruction tuning for format compliance. These operate at the sequence level or rely on training-time intervention. Our approach operates at the token level during inference, requiring no training.

Ontology-aware IE. Schema-informed prompting [Li et al., 2023] provides ontology constraints in the prompt. We show this reduces but does not eliminate violations (§3), motivating decoding-time enforcement. Post-hoc filtering validates outputs against ontology constraints after generation but cannot redirect probability mass to valid alternatives and wastes compute on rejected outputs.

Type-constrained knowledge base completion. Type constraints in knowledge base embedding models [Krompass et al., 2015] enforce domain/range during link prediction. Our semantic layer adapts this concept to autoregressive text generation via dynamic logit masking.

8 Conclusion

We present the first systematic study of ontology constraint violations in LLM-based information extraction, revealing that violations concentrate on specific relation types and that constraint tightness predicts violation rates in small ontologies (ρ=0.85) but not complex ones (ρ=0.14). Grammar-constrained decoding compiled from ontology schemas universally eliminates structural violations (ET→0%, RV→0%) at 14% overhead and zero F1 cost—a no-regret addition to any LLM-based IE pipeline. A semantic constraint layer achieves 94% domain/range violation reduction on a tight ontology (SciERC) but only 35.5% on a complex ontology (DocRED), identifying scaling semantic constraints as an explicit open problem. Our violation landscape measurements, cross-model analyses, and tightness correlation findings provide empirical foundations for future work on ontology-compliant generation.

Limitations. Our study evaluates two datasets and two models; broader validation is needed. The semantic layer introduces cross-field consistency violations (up to 9.9%), an inherent consequence of per-relation independent enforcement. DocRED's 10.19% residual domain/range violation rate under full constraints quantifies the gap between current methods and ontology compliance on complex schemas. All experiments use zero-shot extraction; the interaction between constraint enforcement and fine-tuned models remains unexplored.

A S1 vs. S2 Decoding Strategies

We evaluate an alternative two-pass decoding strategy (S2): Pass 1 extracts entities with grammar-only constraints; Pass 2 generates relations with an input-dependent grammar compiled from Pass 1 entities.

Table 11 shows S2 offers no improvement over the single-pass S1 strategy on either dataset.

S2 slightly reduces CF on SciERC (−1.87pp) but increases it on DocRED (+1.81pp), while DR and F1 are comparable or slightly worse. The 135% latency overhead of S2 (two full inference

Table 11: S1 (single-pass) vs. S2 (two-pass) comparison. S2 shows no advantage on DR, CF, or F1.

Dataset	**S1 DR(%)**	**S2 DR(%)**	**S1 CF(%)**	**S2 CF(%)**	**S1 R-F1(%)**	**S2 R-F1(%)**
SciERC	0.93	0.97	9.9	8.03	8.36	8.03
DocRED	10.19	10.80	2.99	4.80	8.17	7.98

passes) makes it impractical. We retain S1 as the recommended strategy and present S2 as an architectural variant demonstrating framework flexibility.

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